Multispectral satellite imagery for illegal waste material classification

State of the art and thesis direction

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The problem: priority, not just presence

Illegal waste dumping is an environmental crime and a public-health issue.

Agencies such as ARPA cannot inspect everything, they must prioritise by material.

Today's automated pipelines find sites well; they classify presence, not what is there.

Research question

What is the added value of multispectral data, vs. RGB only, for waste material classification from satellite imagery?

Framing distilled from Fraternali et al. 2024 (PoliMi survey, arXiv:2402.09066).

Not all waste is equal: risk = hazard x exposure x magnitude

- The danger is material-bound: inert rubble vs plastics vs asbestos-cement are worlds apart.
- "Material" carries the hazard; the European Waste Catalogue (EWC) flags hazardous codes with *.

  Asbestos = 17 06 05* (mirror-hazardous).
- Priority risk = hazard (material) x exposure (who/what is nearby) x magnitude (how much).
- Goal: turn imagery into an actionable ARPA priority list - computer vision as decision support.

EWC List of Waste (Dec. 2000/532/EC) · Indice di Degrado d.d.g. 13237/2008 · Fazzo et al. 2023

Today's paradigm: RGB deep learning on aerial imagery

VHR aerial / satellite tile (RGB) → CNN or Transformer backbone → pretraining (ImageNet, EO SSL) → two-step fine-tune → binary waste / no-waste.

- Accurate within the geographic context it was trained on.
- Classifies presence, not material.
- Tied to chromatic information only.
- Generalisation collapses outside it, new region or new sensor.

These limits motivate looking at the spectrum.

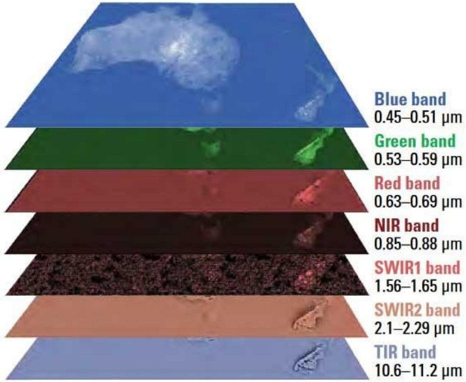
Several recent surveys, e.g. Fraternali et al. 2024.

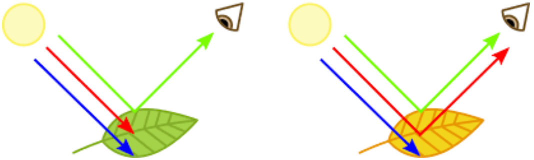
A satellite pixel is a spectrum, not just a colour

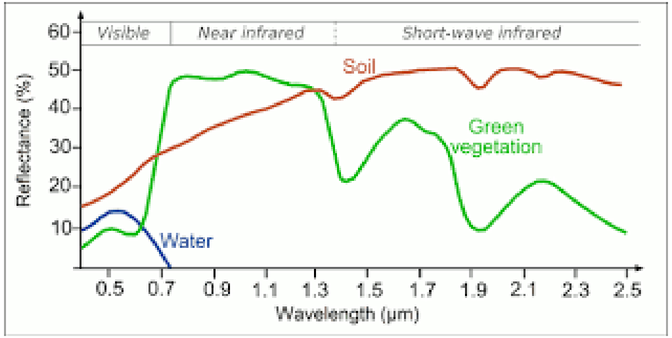
Each pixel is a vector of reflectance values, the material's spectral signature.

- RGB, 3 broad visible bands (colour only).
- Multispectral (MS), 4-15 chosen bands. WorldView-3 records 8 VNIR + 8 SWIR; Pleiades Neo records 6 VNIR.
- Hyperspectral (HSI), hundreds of contiguous narrow bands.

This vector is the raw input from which any classifier reasons.







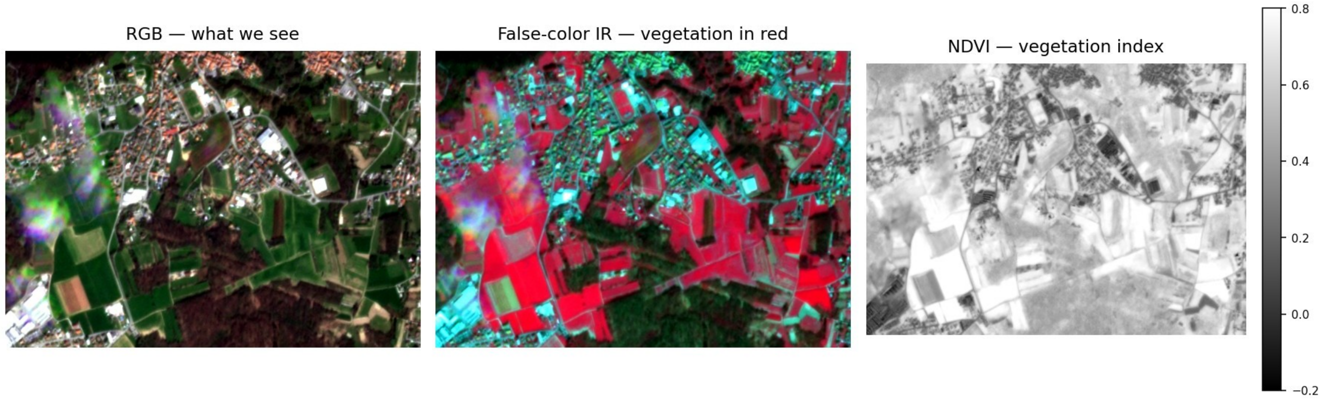
Spectrum schema after USGS splib07a (Kokaly et al. 2017).

From bands to information: three views of the same scene

Same multiband image, many ways to read it.

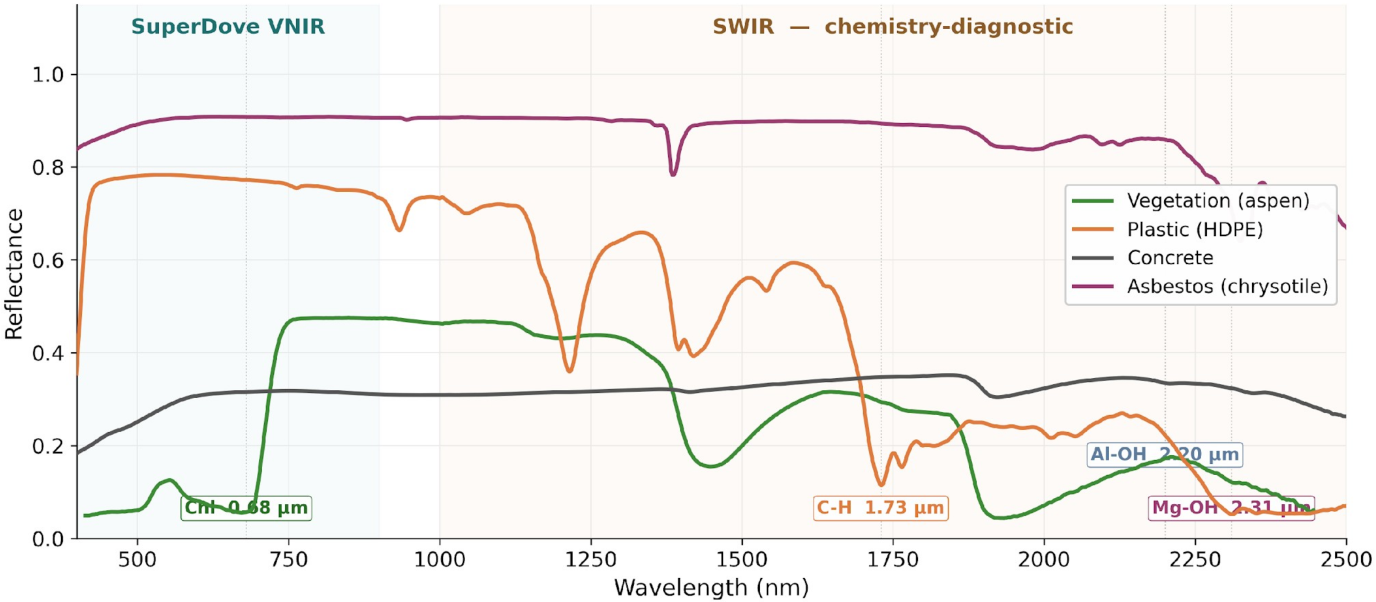
- RGB, what we see.
- False-colour IR, vegetation in red.
- NDVI, vegetation index map.

Material discrimination needs more than RGB.



Planet PSScene, Lombardy 2026 (4-band SR product).

Every material has a spectral fingerprint



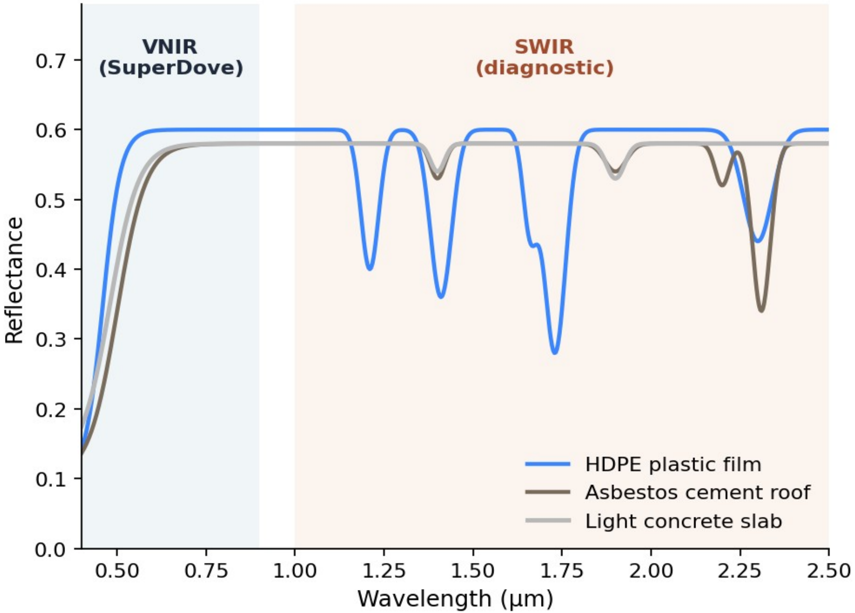
USGS splib07a (Kokaly et al., 2017)

USGS splib07a (Kokaly et al., 2017). Diagnostic features: Cilia 2015, Aguilar 2025.

RGB fails in two distinct ways (Iso-chromaticity)

iso-chromaticity and sub-pixel mixing, both invisible without spectral information.

Same color, different materials



HDPE plastic film

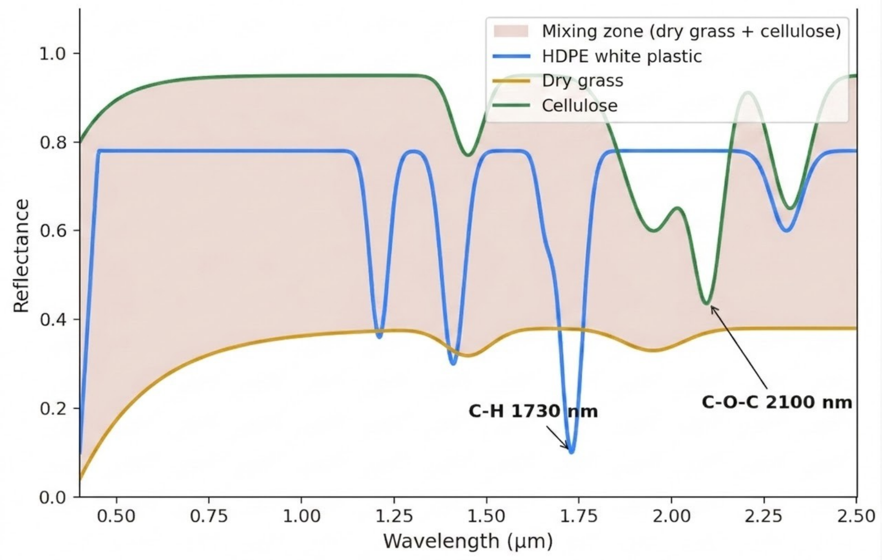
Asbestos-cement roof

Light concrete slab

Tasseron 2021 · Aguilar/Uhrin 2025 · USGS splib07a

Different materials mix into the target's magnitude

- A pixel at 10–30 m rarely contains one pure material.
- A linear mix of two endmembers can land on the spectral magnitude of a third, pure material.
- The mixture cannot be unmixed in RGB, only SWIR features (C-H, C-O-C) keep the materials separable.

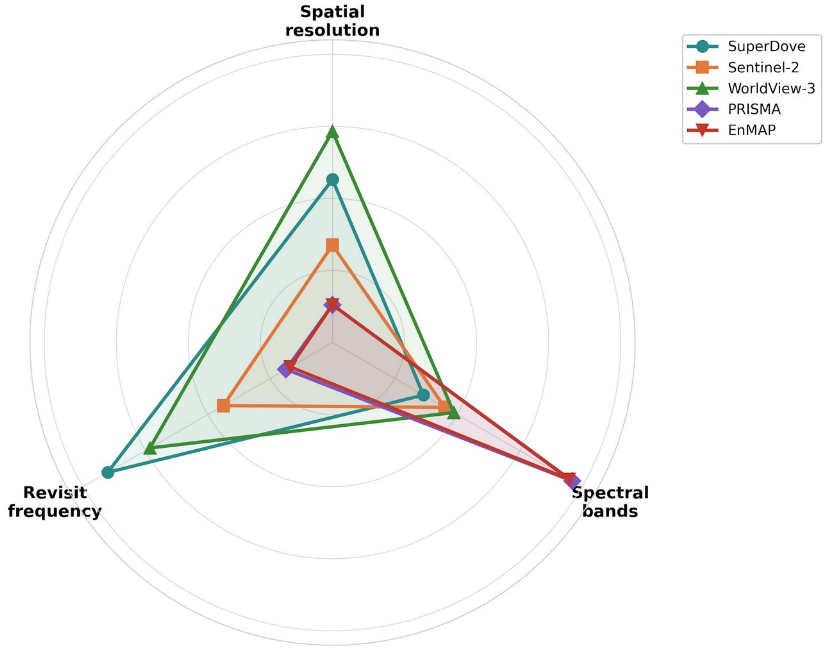


Tasseron 2021 · Aguilar/Uhrin 2025 · USGS splib07a

The sensor trade-off: spatial x spectral x revisit

Three axes, one photon budget. No satellite maximises all three.

- WorldView-3: 1.24 m VNIR + 3.7 m SWIR, 16 bands - the chemistry sensor.
- Pleiades Neo: 1.2 m, 6 VNIR, no SWIR - the VNIR-only cross-sensor.
- Sentinel-2: 10 m, 13 bands, free - too coarse for material.
- SuperDove: 3 m, 8 VNIR, free, near-daily - wide-area screening.
- PRISMA / EnMAP: 30 m, 200+ bands - resolves chemistry but coarse.



Outer ring = better. Axes: Spatial 1/GSD (log) · Bands log10(N) · Revisit 1/days (log).

Specs: Planet, ESA, Maxar, ASI, DLR (mission documents).

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Multispectral at low / medium resolution (10–30 m)

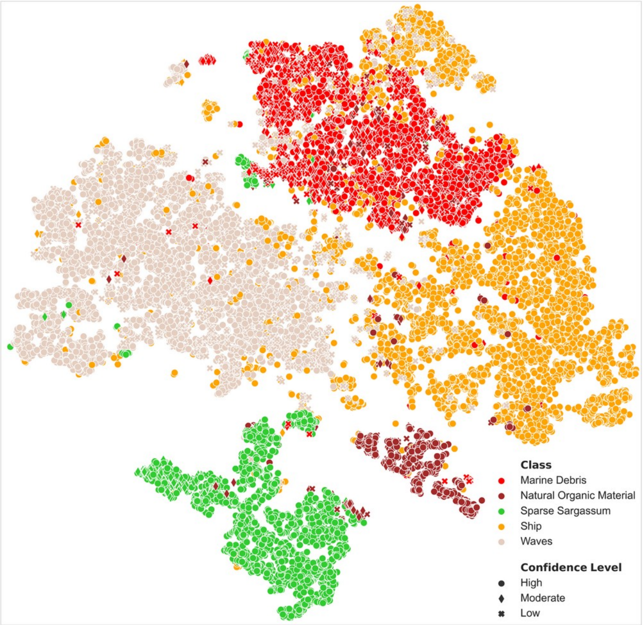
Work	Input	Method	Result	Task
MARIDA (Kikaki 2022)	S-2 13-band @10 m	RF + U-Net, pixel-level	F1 0.79 multi-class (15)	Segmentation (semantic)
Tisza (Magyar 2023)	S-2 + PlanetScope	RF + Plastic Index	Acc. 96 % change detection	Pixel classification
Aharoni-Mack / Shepherd 2025 (asbestos)	EnMAP 230-b @30 m	Cascade 8 classifiers	86 % match, 823 detections	Pixel classification
Sun 2023 (dumpsites, Asia)	S-2 + SuperDove	BCA-Net deep learning	Sensitivity 98 % over 28 cities	Detection

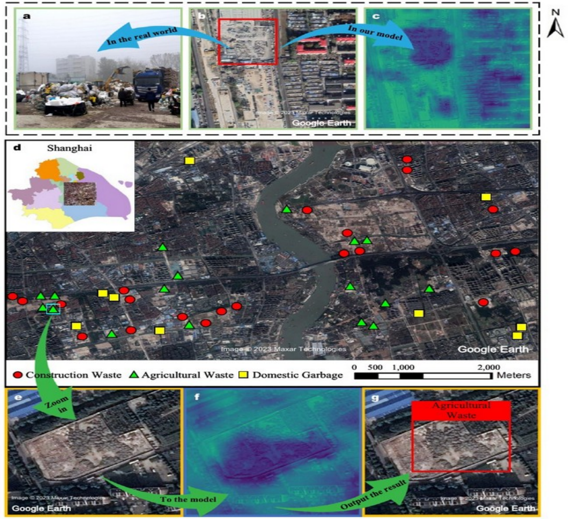
Binary detection works where signal fits 10–30 m pixels. Material classification needs SWIR at adequate resolution (EnMAP), Sentinel-2 has SWIR too, but at 20 m it is not enough.

Kikaki 2022 PLOS ONE · Magyar 2023 arXiv · Shepherd 2025 Sci Rep · Sun 2023 Nature Comms 14:1565.

Low / medium resolution: what they catch, what they miss

- MARIDA (Kikaki 2022), marine debris confused with natural organic matter where pixels mix.
- Sun 2023, dumpsite locations across 28 cities detected, but material identity stays out-of-reach.
- Tisza (Magyar 2023), change detection works; per-tile post-processing still needed.





The chosen data: WorldView-3 + Pleiades Neo

	WorldView-3	Pleiades Neo
VNIR	8 bands @ 1.24 m	6 bands @ 1.2 m
SWIR	8 bands @ 3.7 m	none
Pan	~0.31 m	~0.30 m
Unique bands	Yellow, NIR2	Deep Blue
Role	full ladder incl. SWIR chemistry	VNIR-only cross-sensor axis

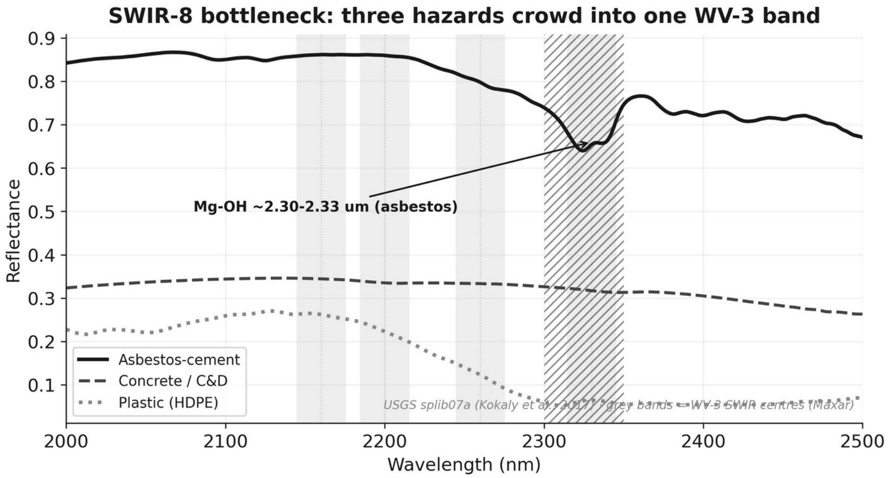
Both sub-metre VHR; access feasible for free via ESA Third-Party Missions (~9-week proposal lead).

WorldView-3 is rare civilian SWIR at high resolution; Pleiades Neo tests how far VNIR-only gets.

Specs: Maxar WorldView-3 · Airbus Pleiades Neo · ESA TPM

Honest caveat: resolution is split between texture and chemistry

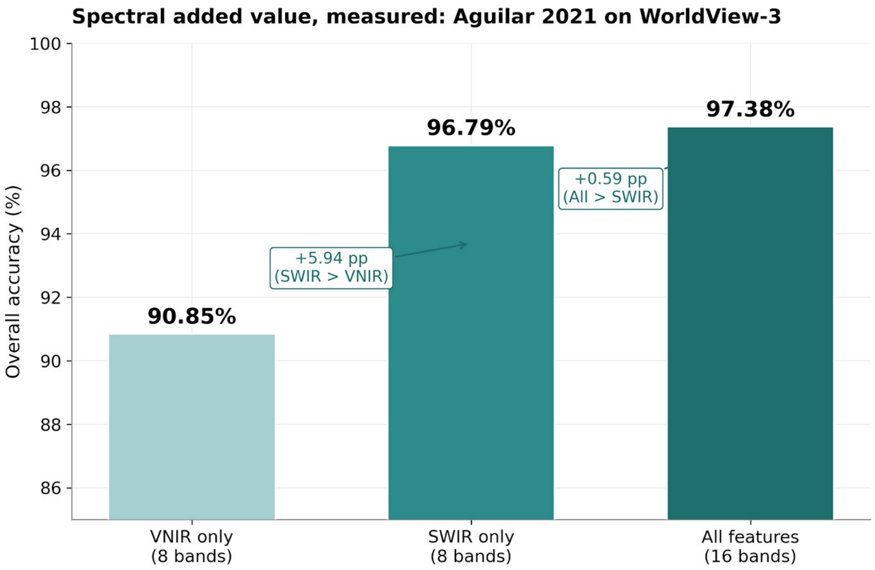
- Spectra are sampled coarsely: VNIR ~1.24 m, SWIR ~3.7 m (~14 m2 pixels).
- The pan band gives fine texture ~0.3 m, but no material info alone.
- A small dump is easier to SEE (texture) than to chemically IDENTIFY (spectra).
- SWIR-8 bottleneck: asbestos 2.32 + concrete 2.34 + plastic 2.31 crowd one band.
- Honest: at 3.7 m + atmosphere, R3 may = R2 - itself a valid finding.



Spectral added value, measured: Aguilar 2021 on WorldView-3

Most-cited band ablation on WV-3 for waste-like classification (greenhouse plastic, n = 14.3 M pixels).

OBIA + Decision Tree, 10-fold CV. The thesis measures exactly this VNIR -> +SWIR gain on waste materials.



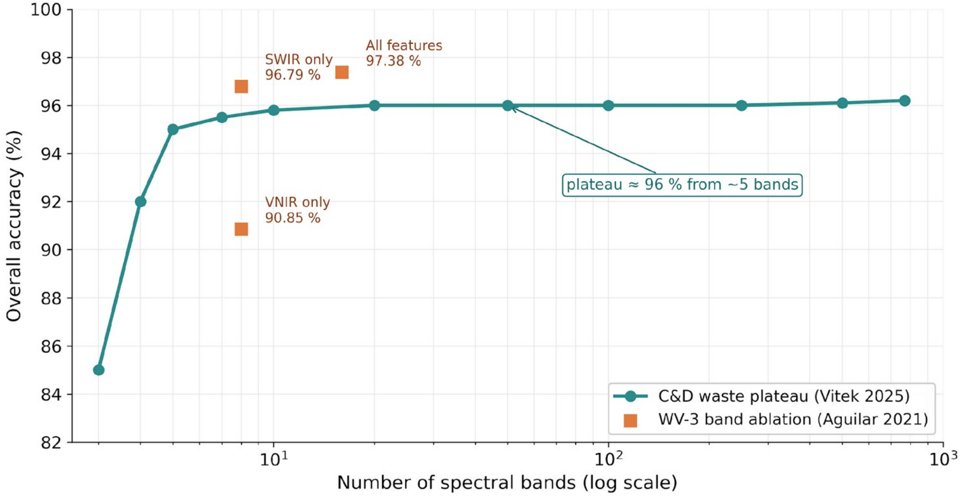
Aguilar, Jiménez-Lao & Aguilar 2021 — Remote Sensing 13(11):2133. OBIA + Decision Tree, 10-fold CV, 14.3 M pixel GT.

Aguilar, Jiménez-Lao & Aguilar 2021, Remote Sensing 13(11):2133. OBIA + DT, 10-fold CV.

More bands ≠ more information, well-chosen few suffice

- Aguilar 2021 (WV-3 plastic): VNIR 90.85 -> All 97.38; SWIR carries the jump.
- Vitek 2025 (C&D waste): RGB + 2 narrowbands = HSI 768 bands.
- Zhou 2021 (WV-3 plastic): 8 SWIR bands separate 3 polymer clusters.

What matters is which bands, not how many. SWIR is the most informative region - and WorldView-3 gives us SWIR; Pleiades Neo (VNIR-only) measures how far we get without it.



Vitek et al. 2025 (CTU Prague, C&D waste) — Stage-1/2 narrowband selection on HSI 425–975 nm · Aguilar 2021 verified above.

Asbestos: direct precedents to benchmark against

- Saba 2026 (WV-3 VNIR, 8 bands): 32 classifiers; Fine-KNN Macro-F1 97.6% - strong VNIR-only upper bound. [paywalled]
- Bonifazi 2026 (WV-3, 16 bands, Max-Likelihood): AC roofs F1 0.87 (P 0.81 / R 0.92); 25,319 buildings; multi-temporal.
- Shepherd 2025 (EnMAP, 230 bands @ 30 m): ACE 91.4% OA / F1 91.2%; 86% field match - but urban mixed pixels mimic asbestos.

Together: VNIR can detect AC; SWIR + hyperspectral add robustness on clean / ambiguous roofs.

Saba 2026 J. Hazard. Mater. · Bonifazi 2026 Geomatics 6(3):41 · Shepherd 2025 Sci. Rep.

Where each hazard becomes separable: feature -> band

Where each hazard becomes separable: diagnostic feature to band

	Pleiades Neo (VNIR-only)			WorldView-3 adds SWIR			
	RGB	Red Edge	NIR	SWIR 1.2um	SWIR 1.73um	SWIR 2.2um	SWIR 2.33um
Asbestos (Mg-OH)						weak	strong
Plastics (C-H)				strong	strong	weak	weak
Concrete/C&D (carbonate)						strong	strong
Slag/rust (Fe-oxide)		weak	strong				
AC weathering (moss/lichen)		strong	weak				

USGS splib07a · Aguilar 2021/2025 · Cilia 2015

Pleiades Neo reaches up to NIR; WorldView-3 adds the SWIR chemistry · USGS splib07a · Aguilar · Cilia 2015

Foundation models for Earth Observation (EO): a new lever

2024–2025: a wave of pretrained backbones for EO.

Most are pretrained at 10–30 m on Sentinel-2 / HLS, naïve transfer to 3 m SuperDove is not guaranteed.

Two routes forward

- Sensor-agnostic models (DOFA, AnySat), accept arbitrary band sets.
- Parameter-efficient adapters (DEFLECT), freeze the backbone, add small heads (<1 % parameters).

Open question: does pretrain at 10–30 m transfer well to 3 m?

FM-RS survey 2024 · Xiong 2024 · Astruc 2024 · Thoreau 2025 · Szwarcman 2024.

DOFA: a band-agnostic backbone, the candidate to start from

Dynamic One-For-All (Xiong et al. 2024).

- A hypernetwork generates patch-embedding weights from each band's central wavelength.
- One backbone handles WV-3 (16 b), Pleiades Neo (6 b), Sentinel-2 (13 b), or HSI.
- So RGB vs VNIR vs SWIR becomes a fair comparison, not apples-to-oranges.
- Natural fit for the band-ablation study; pairable with a DEFLECT adapter (<1% params).

To verify: does pretraining at 10-30 m close the gap to ~1.2 m VHR?

Xiong et al. 2024 (arXiv:2403.15356) · Thoreau et al. 2025 (DEFLECT, ICCV).

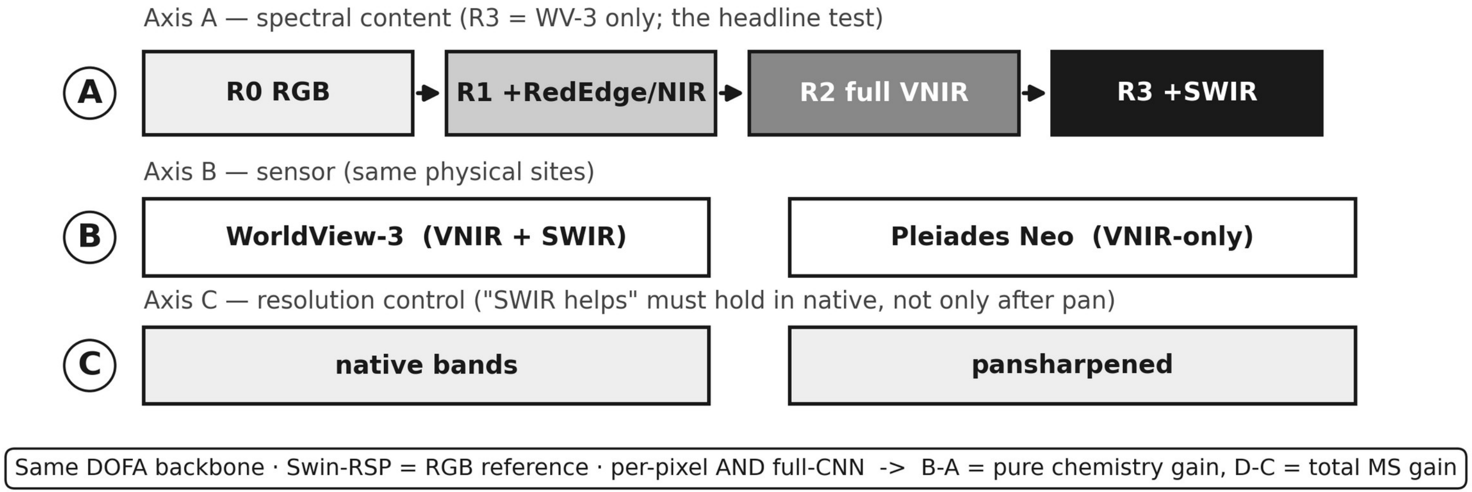
Gaps in the literature, and the test that matters

- Material-level labels are missing: no public VHR waste dataset annotates the material.
- No peer-reviewed Pleiades-Neo material benchmark exists.
- RGB pipelines conflate iso-colour, chemically distinct materials.
- No pipeline goes from spectra to a regulatory hazard class (material -> EWC -> risk).
- The object-vs-material split is unaddressed; generalizzazione (region/sensor/time) rarely shown.

The test that matters: how much does multiband beat RGB for material, and where does it stop?

Gap list distilled from Fraternali et al. 2024 (PoliMi survey, arXiv:2402.09066).

The experiment: a 3-axis band ablation



Design: loop_prof_sota/04_experimental_design.md

Per-pixel AND full-CNN: B-A = pure chemistry gain, D-C = total MS gain · loop_prof_sota/04_experimental_design.md

Generalization as a first-class axis (expected, to be measured)

	In-domain	Cross-region	Cross-sensor
R0 RGB	baseline	-5.1% (Gibellini)	ports trivially
R2 full VNIR	+ small	gap narrows	needs harmonization
R3 +SWIR	+ chemistry	gap narrows most	widens unless SBAF

Whether added bands narrow or widen the generalization gap is publishable either way - report ID-OOD dF1 per cell.

Cross-region -5.1%: Gibellini 2025 · cross-sensor: GeoCrossBench - cell values EXPECTED

From material to risk tier

- Map predicted material -> EWC hazard code -> Indice di Degrado / PRAL x exposure x magnitude -> ARPA priority tier.
- Output = a ranked intervention list; every step transparent and auditable.

  Indice di Degrado (<=25 / 26-44 / >=45 -> action windows) is NOT in the public WFS

  -> estimated remotely (SWIR Mg-OH depth + VNIR weathering) = a thesis contribution.
- Exposure grounded in Fazzo et al. (residential-proximity mesothelioma risk); magnitude from site area.

EWC 17 06 05* · Indice di Degrado d.d.g. 13237/2008 · Fazzo et al. 2023

Proposed direction: from controlled pilot to public benchmark

Phase 1 - asbestos spectral pilot (run first, in full)

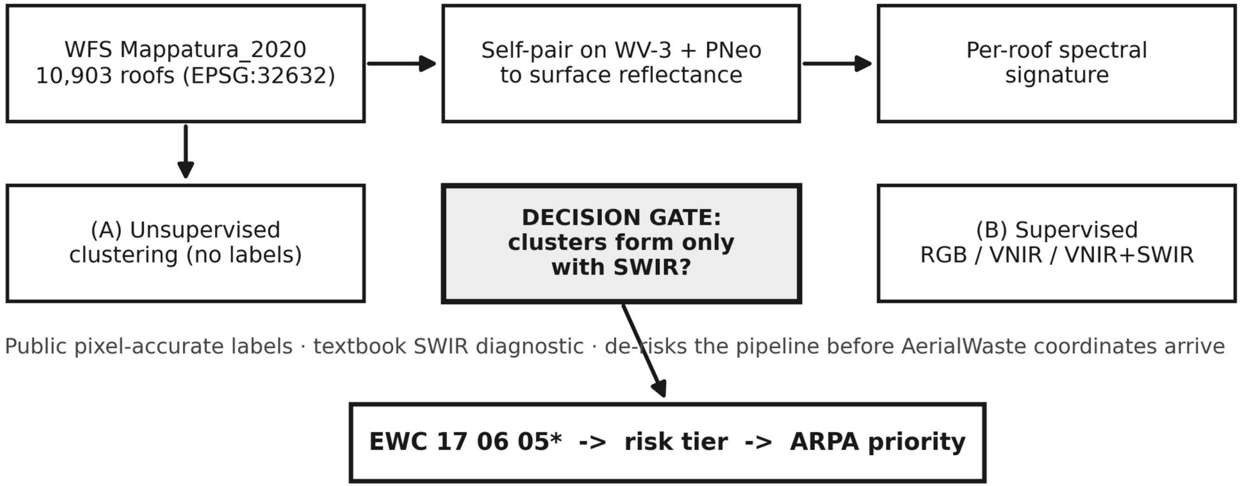
- Public GT: Lombardy WFS Mappatura 2020 (10,903 roofs). Textbook SWIR diagnostic (Mg-OH ~2.3 um).
- Self-pair on WV-3 + PNeo -> surface reflectance -> per-roof signatures.
- Unsupervised clustering (no labels) -> decision gate: do AC clusters form only with SWIR?
- Then map material -> EWC 17 06 05* -> risk tier -> ARPA priority.

Phase 2 - multispectral waste benchmark

- Co-register AerialWaste with the VHR archive; 3-axis ablation (R0-R3 x WV-3/PNeo x native/pansharp).
- DOFA backbone; cross-region / cross-sensor / cross-time splits.

Pilot GT: Mappatura 2020 (Lombardia WFS) · Backbone: DOFA (Xiong 2024) · Baseline ref.: Gibellini 2025

The asbestos pilot: the immediately-feasible demonstrator



GT: Lombardia WFS Mappatura_2020 · physics: chrysotile Mg-OH 2.30-2.33um (splib07a)

Public pixel-accurate labels + textbook SWIR diagnostic (Mg-OH 2.30-2.33 um) - de-risks the whole pipeline

What is novel - and what to confirm with Thomas

Genuinely novel

- First to MEASURE RGB vs VNIR vs SWIR added value for waste material at VHR.
- First WorldView-3 vs Pleiades Neo head-to-head for this task.
- First to connect spectra -> EWC hazard -> ARPA risk tier; confidence-aware.

To confirm with Thomas

- Trigger the ESA TPM proposal now (~9-wk lead)? AOI over Lombardia?
- Risk-chain depth: full Indice di Degrado, or stop at material + EWC?
- SuperDove kept as wide-area layer, or dropped? · Backbone DOFA + Swin-RSP? · AerialWaste coordinate bridge?

Open dependencies: loop_prof_sota/00_LOOP_LOG.md